Less environmental impact. Optical communication does not produce the same level of noise pollution as acoustic communication, which can be disruptive to marine life. It is a more environmentally friendly option in this regard.

Applications

The potential applications of underwater optical communication are vast and include:

Subsea exploration. Oceanographers, marine biologists, and other scientists often use underwater optical communication to transmit data from sensors, cameras, and other instruments on underwater vehicles, remotely operated vehicles, and autonomous underwater vehicles. This enables real-time monitoring and data collection from deep-sea environments.

Environmental monitoring. Underwater optical communication is employed for environmental monitoring, including the measurement of water quality, temperature, and other parameters in oceans, lakes, and rivers. Transmitting this data in real time is important for assessing environmental conditions.

Military and defence. Underwater optical communication is employed by the military for secure and high-bandwidth communication between submarines, divers, and surface vessels. It helps maintain situational awareness and supports various naval operations.

Underwater infrastructure. In the oil and gas industry, underwater optical communication is used for tasks such as controlling remotely operated vehicles and transmitting data between underwater sensors and surface facilities. This is crucial for the maintenance and operation of subsea infrastructure.

Underwater imaging and surveillance. Optical communication is used in underwater imaging and surveillance systems to transmit high-definition video and images in real time. This is valuable for underwater security, search and rescue operations, and underwater archaeology.

Standardisation activities

Standard development organisations like and International Electrotechnical Commission (IEC), Institute of Electrical and Electronics Engineers (IEEE), and International Telecommunication Union (ITU) are working on various aspects of UWOC.

IEC 62628. This standard provides guidelines and requirements for optical wireless communication systems used in underwater applications, including communication between underwater vehicles, sensors, and other devices. It covers aspects such as modulation schemes, transmission protocols, and system performance.

IEEE 802.15.7. This standard focuses on visible light communication systems, which can be adapted for underwater optical communication. It defines the physical layer and media access control layer specifications for short-range optical wireless communication.

ITU-T G.6980. While not specifically designed for underwater communication, this ITU standard defines the grid for wavelength division multiplexing applications. Some underwater optical communication systems may leverage wavelength division multiplexing technology for increased data rates and channel capacity.

Challenges in UWOC

UWOC is a growing technology with boundless potential. Yet, with this growth comes a set of challenges:

The range limitations. Water, the very medium we seek to communicate through, poses a formidable challenge. Its tendency to absorb and scatter light limits the range of underwater optical communication systems. Researchers are tirelessly working on extending these limits through signal processing advancements and innovative beam-forming techniques.

Murky waters. The clarity of water varies significantly, and impurities and turbidity can cloud optical signals. Adaptable and robust signal processing algorithms are vital to combat these environmental factors.

The dance of precision. Maintaining the delicate alignment between transmitting and receiving devices in dynamic underwater environments is akin to a ballet. The constant motion of water necessitates sophisticated tracking and beam-forming systems.

The cost of exploration. Implementing underwater optical communication can be expensive, especially for deep-sea applications. Balancing the costs while pushing the boundaries of technology requires judicious investment.

Bioluminescence and ambient light. Bioluminescence and ambient light present significant obstacles for UWOC systems in their efforts to enhance data transmission. Bioluminescence is the capacity of specific organisms to generate light via a chemical process. Ambient light arises from sunlight scattering and bending as it passes through the ocean's layers.

Underwater optical communication represents a transformative technology for exploring and exploiting the depths of our oceans. Its high data rates, low latency, and secure transmission open up new possibilities for underwater research, exploration, and applications. As researchers continue to overcome the challenges associated with this technology, we can expect to see even more innovative applications in the future, shedding light on the mysteries of the deep. EFY